

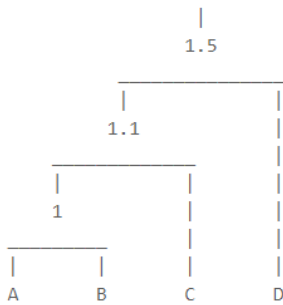
Esercizio

Utilizzare l'algoritmo "Hierarchical Agglomerative Clustering" con metrica *single link* per eseguire il clustering dei seguenti punti:

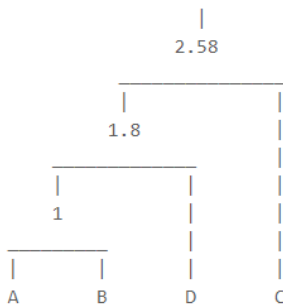
$A = [0 \ 1]$; $B = [1 \ 1]$; $C = [2.1 \ 1]$; $D = [0 \ 2.5]$;

Selezionare il Dendrogramma corrispondente:

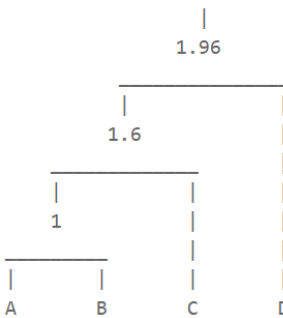
A



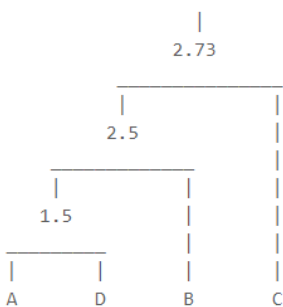
B



C



D



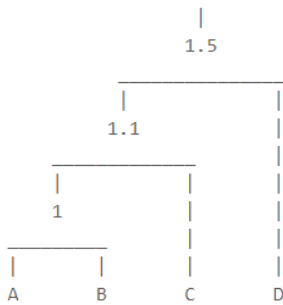
Esercizio

Utilizzare l'algoritmo "Hierarchical Agglomerative Clustering" con metrica *Complete link* per eseguire il clustering dei seguenti punti:

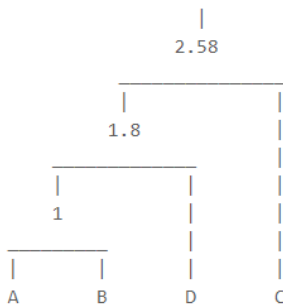
$A = [0 \ 1]$; $B = [1 \ 1]$; $C = [2.1 \ 1]$; $D = [0 \ 2.5]$;

Selezionare il Dendrogramma corrispondente:

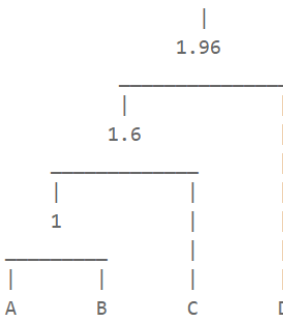
A



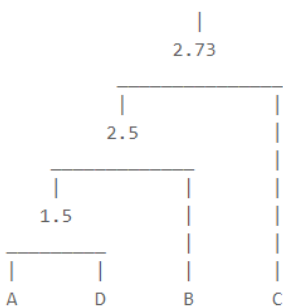
B



C



D



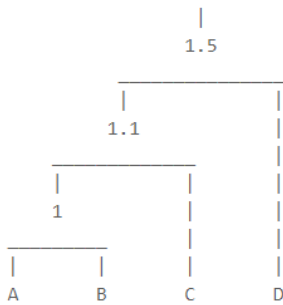
Esercizio

Utilizzare l'algoritmo "Hierarchical Agglomerative Clustering" con metrica *Average link* per eseguire il clustering dei seguenti punti:

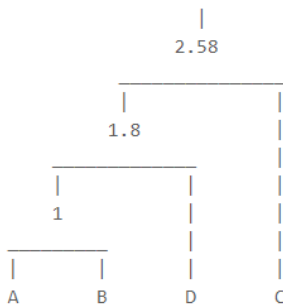
$A = [0 \ 1]$; $B = [1 \ 1]$; $C = [2.1 \ 1]$; $D = [0 \ 2.5]$;

Selezionare il Dendrogramma corrispondente:

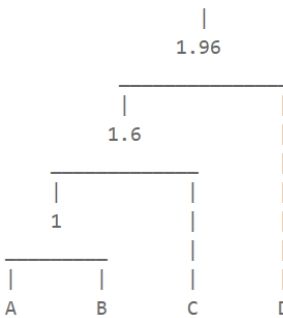
A



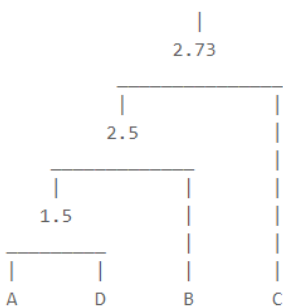
B



C

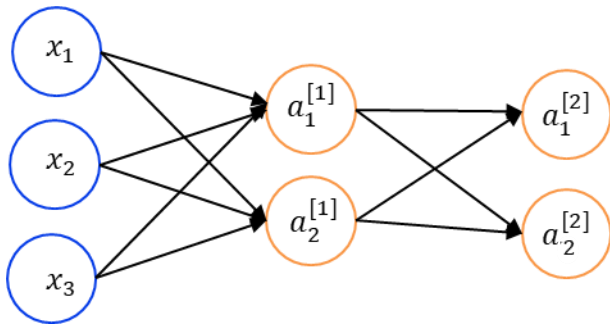


D



Esercizio

Supponiamo di avere la seguente rete neurale:



Supponiamo inoltre, dopo aver addestrato la rete, di avere i seguenti parametri

$$W^{[1]} = \begin{bmatrix} 4 & -3 & 2 \\ -6 & -2 & 0 \end{bmatrix}; b^{[1]} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}; W^{[2]} = \begin{bmatrix} 2 & 1 \\ 2 & -1 \end{bmatrix}; b^{[2]} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Funzione di attivazione di $a^{[1]}$: Sigmoid

Funzione di attivazione di $a^{[2]}$: Sigmoid

Calcolare l'output della rete per il vettore $x = [2 \ 3 \ 1]^T$

Risposta: **[0.269 0.731]**

Cenni di Soluzione:

$$z_1^{[1]} = W_{11}^{[1]} x_1 + W_{12}^{[1]} x_2 + W_{13}^{[1]} x_3 + b_1^{[1]} = (4 * 2 - 3 * 3 + 2 * 1 + 3) = 4$$

$$a_1^{[1]} = g(z_1^{[1]}) = g(4) = 0.982$$

$$z_2^{[1]} = W_{21}^{[1]} x_1 + W_{22}^{[1]} x_2 + W_{23}^{[1]} x_3 + b_2^{[1]} = (-6 * 2 - 2 * 3 + 0 * 1 + 2) = -16$$

$$a_2^{[1]} = g(z_2^{[1]}) = g(-16) = 0$$

$$z_1^{[2]} = W_{11}^{[2]} a_1^{[1]} + W_{12}^{[2]} a_2^{[1]} + b_1^{[2]} = (2 * 0.982 + 1 * 0 + 1) = 2.964$$

$$z_2^{[2]} = W_{21}^{[2]} a_1^{[1]} + W_{22}^{[2]} a_2^{[1]} + b_2^{[2]} = (2 * 0.982 - 1 * 0 + 2) = 3.964$$

$$a_1^{[2]} = \frac{e^{2.964}}{e^{2.964} + e^{3.964}} = 0.269$$

$$a_2^{[2]} = \frac{e^{3.964}}{e^{2.964} + e^{3.964}} = 0.731$$

Where $g(z) = \frac{1}{1+e^{-z}}$ is the Sigmoid function.

Nota, possibili varianti: Funzione di attivazione ReLU